

TASK RECORD

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TASK: Basic Linux commands, User Management, Groups and Vim

Task Description:

The objective of this task is to learn and practice basic Linux user management and groups and vim.

TASK:

Linux is an operating system that can be controlled using commands in the terminal. In our practical sessions, I learned basic Linux commands first and then practiced file handling, Vim editor commands, user management, group management, permissions and process management. This document contains the commands I practiced and my understanding of their uses.

OBJECTIVES:

- To understand how to use the Linux terminal.
- To practice basic Linux commands.
- To create, view, edit, copy, move and delete files and directories.
- To understand users, passwords and groups.
- To learn how to change file permissions using `chmod`.
- To practice Vim editor commands.
- To understand processes and the commands.

1.BASIC LINUX COMMANDS:

- **Pwd** : Shows the present working directory.
- **Ls** : Lists files and directories in the current directory.
- **ls -l** : Shows detailed information such as permissions, owner, group, size and time.
- **ls -a** : Shows hidden files also.
- **ls -lart** : Shows detailed information including hidden files, arranged by modification time.
- **Mkdir(folder name)** : Creates a new folder named .

- **cd(folder name)** : Enters the ubuntu-practice folder.
- **touch file1.txt** : Creates an empty file named file1.txt.
- **touch file2.txt** : Creates an empty file named file2.txt.
- **cp file1.txt copyfile1.txt** : Creates a copy of file1.txt with the name copyfile1.txt.
- **mv file2.txt myfile.txt** : Renames file2.txt to myfile.txt.
- **mv myfile.txt notes/** : Moves myfile.txt into the notes folder.
- **rm copy_file1.txt** : Deletes the specified file. The exact filename must exist.
- **rm -r notes** : Deletes the notes folder and its contents.
- **rm -rf(folder name)** : Forcefully removes a folder and its contents. Use carefully because it does not ask for confirmation.

2. MOVING A FILE INTO A FOLDER:

- Before using **mv myfile.txt notes/**, the notes folder must exist. If it does not exist, create it first.
- **mkdir notes** : Creates the notes folder.
- **mv myfile.txt notes/** : Moves myfile.txt into notes.
- **ls** : Checks the current folder.
- **ls notes** : Checks the contents of notes.

3. DELETING FILES AND FOLDERS:

- The **rm** command is used to delete files. The **rm -r** command is used to delete a directory and its contents. The **-f** option forces removal, so it should be used carefully.
- **rm copyfile1.txt** : Deletes the copied file.
- **ls** : Checks that the file has been removed.
- **rm -r notes** : Deletes the notes folder and its contents.
- **ls** : Checks that notes is removed.

4.VIM EDITOR COMMANDS:

- Vim is a terminal text editor. I practiced opening a file, entering insert mode, typing text, saving the file and quitting Vim.
- **Vim -info.txt** : Opens the file or creates it if it does not exist.
- **I** : Enters Insert mode so that text can be typed.
- **Esc** : Leaves Insert mode.
- **:w** : Saves the file.
- **:q** : Quits Vim.
- **:wq** : Saves and quits Vim.
- **:q!** : Quits without saving changes.
- **:w!** : Forces a save when allowed.
- **:wq!** : Saves forcefully and quits when allowed.

- Example Vim practice:
- `vim vikash-info.txt`
- Press `i`
- Type: Hello Vikash
- Type: How are you
- Press `Esc`
- Type `:wq`
- Press `Enter`
- `cat vikash-info.txt`
- `cd /home/vikasini`: Moves to the specified directory.

5.USER MANAGEMENT COMMANDS:

- `sudo useradd (name)` : Creates a user name.
- `sudo useradd -m` : Creates user name and creates the user's home directory.
- `sudo passwd` : Sets or changes the password .
- `id` : Shows UID, GID and groups name.
- `cat /etc/passwd` : Displays Linux user account entries.
- `grep name /etc/passwd` : Searches for the user entry.
- `sudo userdel` : Deletes the user account.
- `sudo userdel -r` : Deletes the user and the user's home directory.

6.CHANGING A USER'S HOME DIRECTORY:

- I practiced changing the home directory of vikasini2. The `-d` option specifies the new home directory and `-m` moves the existing home directory contents.
- `sudo mkdir /home/newhome`
- `sudo usermod -d /home/newhome -m vikasini2.`

7.RETURNING TO MY MAIN HOME DIRECTORY:

- `cd /home/vikasini`
- `pwd`
- `cd ~`
- `pwd`

8.GROUP MANAGEMENT COMMANDS:

- `sudo groupadd` : Creates a group name.
- `groups name` : Shows the groups to which belongs.
- `id name` : Shows UID, primary group and supplementary groups.
- `sudo usermod -aG name` : Adds one and two as a supplementary group.
- `sudo usermod -g vikash student` : Changes the primary group of student to vikash.
- `sudo groupdel name` : Deletes the sruthi group when it is not being used as a primary group.
- `sudo groupmod -n classmates students` : Renames the students group to classmates.

9.ROOT USER COMMANDS:

- **sudo -l** : Opens a root shell.
- **sudo su** : Switches to the root user.
- **Whoami** : Confirms the current user; it should show root.
- **Pwd** : Shows the current directory; root's home is normally /root.
- **cd /** : Moves to the root of the Linux filesystem.
- **Ls** : Lists directories and files under the current location.
- **cd /root** : Moves to the root user's home directory when permission allows.
- **Exit** : Returns from the root shell.

OUTPUT:

```
vikasini@LAPTOP-DPLUR7N6:~$ pwd
/home/vikasini
vikasini@LAPTOP-DPLUR7N6:~$ ls
chmod_practice  new-folder  ubuntupractice
vikasini@LAPTOP-DPLUR7N6:~$ ls -l
chmod_practice
new-folder
ubuntupractice
vikasini@LAPTOP-DPLUR7N6:~$ ls -a
.  .bash_history  .bashrc  .gitconfig  .motd_shown  .sudo_as_admin_successful  new-folder
.. .bash_logout  .cache  .lesshst  .profile  chmod_practice  ubuntupractice
vikasini@LAPTOP-DPLUR7N6:~$ ls -la
total 52
drwxr-x--- 6 vikasini vikasini 4096 Aug  7 18:18 .
drwxr-xr-x 5 root      root      4096 Aug 29 18:09 ..
-rw----- 1 vikasini vikasini 4124 Aug 29 18:12 .bash_history
-rw-r--r-- 1 vikasini vikasini  220 Jul 26 07:45 .bash_logout
-rw-r--r-- 1 vikasini vikasini 3771 Jul 26 07:45 .bashrc
drwx----- 2 vikasini vikasini 4096 Jul 26 07:45 .cache
-rw-r--r-- 1 vikasini vikasini   63 Jul 26 08:46 .gitconfig
-rw----- 1 vikasini vikasini   20 Aug  7 18:18 .lesshst
-rw-rw-r-- 1 vikasini vikasini    0 Sep  1 15:39 .motd_shown
-rw-r--r-- 1 vikasini vikasini   807 Jul 26 07:45 .profile
-rw-r--r-- 1 vikasini vikasini    0 Jul 26 08:39 .sudo_as_admin_successful
drwxr-xr-x 2 vikasini vikasini 4096 Aug  3 16:23 chmod_practice
drwxr-xr-x 2 vikasini vikasini 4096 Jul 29 16:18 new-folder
drwxr-xr-x 3 vikasini vikasini 4096 Jul 26 11:13 ubuntupractice
vikasini@LAPTOP-DPLUR7N6:~$ mkdir practice
vikasini@LAPTOP-DPLUR7N6:~$ ls
chmod_practice  new-folder  practice  ubuntupractice
vikasini@LAPTOP-DPLUR7N6:~$ cd practice
vikasini@LAPTOP-DPLUR7N6:~/practice$ pwd
/home/vikasini/practice
vikasini@LAPTOP-DPLUR7N6:~/practice$ ls
vikasini@LAPTOP-DPLUR7N6:~/practice$ touch day1.txt
vikasini@LAPTOP-DPLUR7N6:~/practice$ touch day2.txt
vikasini@LAPTOP-DPLUR7N6:~/practice$ ls
```

```
vikasini@LAPTOP-DPLUR7N6: X + v
vikasini@LAPTOP-DPLUR7N6:~/practice2$ ls
day1.txt day2.txt
vikasini@LAPTOP-DPLUR7N6:~/practice2$ cp day1.txt copyday2.txt
vikasini@LAPTOP-DPLUR7N6:~/practice2$ ls
copyday2.txt day1.txt day2.txt
vikasini@LAPTOP-DPLUR7N6:~/practice2$ mv day2.txt myfile.txt
vikasini@LAPTOP-DPLUR7N6:~/practice2$ ls
copyday2.txt day1.txt myfile.txt
vikasini@LAPTOP-DPLUR7N6:~/practice2$ mkdir notes
vikasini@LAPTOP-DPLUR7N6:~/practice2$ mv mylife.txt notes/
mv: cannot stat 'mylife.txt': No such file or directory
vikasini@LAPTOP-DPLUR7N6:~/practice2$ mv myfile.txt notes/
vikasini@LAPTOP-DPLUR7N6:~/practice2$ ls
copyday2.txt day1.txt notes
vikasini@LAPTOP-DPLUR7N6:~/practice2$ cd notes
vikasini@LAPTOP-DPLUR7N6:~/practice2/notes$ ls
myfile.txt
vikasini@LAPTOP-DPLUR7N6:~/practice2/notes$ cd ..
vikasini@LAPTOP-DPLUR7N6:~/practice2$ pwd
/home/vikasini/practice2
vikasini@LAPTOP-DPLUR7N6:~/practice2$ cd ~/practice2
vikasini@LAPTOP-DPLUR7N6:~/practice2$ ls
copyday2.txt day1.txt notes
vikasini@LAPTOP-DPLUR7N6:~/practice2$ rm copyday1.txt
rm: cannot remove 'copyday1.txt': No such file or directory
vikasini@LAPTOP-DPLUR7N6:~/practice2$ rm notes/myfile.txt
vikasini@LAPTOP-DPLUR7N6:~/practice2$ ls
copyday2.txt day1.txt notes
vikasini@LAPTOP-DPLUR7N6:~/practice2$ rmdir notes
vikasini@LAPTOP-DPLUR7N6:~/practice2$ ls
copyday2.txt day1.txt
```

```
vikasini@LAPTOP-DPLUR7N6: X + v
vikasini@LAPTOP-DPLUR7N6:~/practice2$ ls
day1.txt day2.txt
vikasini@LAPTOP-DPLUR7N6:~/practice2$ cp day1.txt copyday2.txt
vikasini@LAPTOP-DPLUR7N6:~/practice2$ ls
copyday2.txt day1.txt day2.txt
vikasini@LAPTOP-DPLUR7N6:~/practice2$ mv day2.txt myfile.txt
vikasini@LAPTOP-DPLUR7N6:~/practice2$ ls
copyday2.txt day1.txt myfile.txt
vikasini@LAPTOP-DPLUR7N6:~/practice2$ mkdir notes
vikasini@LAPTOP-DPLUR7N6:~/practice2$ mv mylife.txt notes/
mv: cannot stat 'mylife.txt': No such file or directory
vikasini@LAPTOP-DPLUR7N6:~/practice2$ mv myfile.txt notes/
vikasini@LAPTOP-DPLUR7N6:~/practice2$ ls
copyday2.txt day1.txt notes
vikasini@LAPTOP-DPLUR7N6:~/practice2$ cd notes
vikasini@LAPTOP-DPLUR7N6:~/practice2/notes$ ls
myfile.txt
vikasini@LAPTOP-DPLUR7N6:~/practice2/notes$ cd ..
vikasini@LAPTOP-DPLUR7N6:~/practice2$ pwd
/home/vikasini/practice2
vikasini@LAPTOP-DPLUR7N6:~/practice2$ cd ~/practice2
vikasini@LAPTOP-DPLUR7N6:~/practice2$ ls
copyday2.txt day1.txt notes
vikasini@LAPTOP-DPLUR7N6:~/practice2$ rm copyday1.txt
rm: cannot remove 'copyday1.txt': No such file or directory
vikasini@LAPTOP-DPLUR7N6:~/practice2$ rm notes/myfile.txt
vikasini@LAPTOP-DPLUR7N6:~/practice2$ ls
copyday2.txt day1.txt notes
vikasini@LAPTOP-DPLUR7N6:~/practice2$ rmdir notes
vikasini@LAPTOP-DPLUR7N6:~/practice2$ ls
copyday2.txt day1.txt
```

```
vikasini@LAPTOP-DPLUR7N6: ~$ cd ..
vikasini@LAPTOP-DPLUR7N6: ~$ man ls
vikasini@LAPTOP-DPLUR7N6: ~$ man pwd
vikasini@LAPTOP-DPLUR7N6: ~$ cd ~/practice
vikasini@LAPTOP-DPLUR7N6: ~/practice$ ls
copyday1.txt  day1.txt  myfile.txt
vikasini@LAPTOP-DPLUR7N6: ~/practice$ nano day1.txt
vikasini@LAPTOP-DPLUR7N6: ~/practice$ cat day1.txt
Hello Vikasini
I am learning Linux commands
vikasini@LAPTOP-DPLUR7N6: ~/practice$ ls
copyday1.txt  day1.txt  myfile.txt
vikasini@LAPTOP-DPLUR7N6: ~/practice$ vim myfile.txt
vikasini@LAPTOP-DPLUR7N6: ~/practice$ cat myfile.txt
Hello this is vikasini
"I'm running all the linux commands Sir taught us".
vikasini@LAPTOP-DPLUR7N6: ~/practice$ cd ..
vikasini@LAPTOP-DPLUR7N6: ~$ sudo apt update
[sudo] password for vikasini:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [981 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1232 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [288 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [17.7 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1690 kB]
```

```
vikasini@LAPTOP-DPLUR7N6: ~$ chmod g
chmod: missing operand after 'g'
Try 'chmod --help' for more information.
vikasini@LAPTOP-DPLUR7N6: ~$ chmod g
chmod: missing operand after 'g'
Try 'chmod --help' for more information.
vikasini@LAPTOP-DPLUR7N6: ~$ chmod g+w permission.txt
vikasini@LAPTOP-DPLUR7N6: ~$ ls -lart
total 16
-rw-r--r-- 1 vikasini vikasini 0 Sep 1 15:48 copyday1.txt
-rw-r--r-- 1 vikasini vikasini 44 Sep 2 03:21 day1.txt
-rw-r--r-- 1 vikasini vikasini 75 Sep 2 03:27 myfile.txt
drwxr-x--- 9 vikasini vikasini 4096 Sep 2 03:27 ..
-rwxr--r-- 1 vikasini vikasini 0 Sep 2 03:47 permission.txt
drwxr-xr-x 2 vikasini vikasini 4096 Sep 2 03:47 .
vikasini@LAPTOP-DPLUR7N6: ~$ chmod o-r permission.txt
vikasini@LAPTOP-DPLUR7N6: ~$ ls -lart
total 16
-rw-r--r-- 1 vikasini vikasini 0 Sep 1 15:48 copyday1.txt
-rw-r--r-- 1 vikasini vikasini 44 Sep 2 03:21 day1.txt
-rw-r--r-- 1 vikasini vikasini 75 Sep 2 03:27 myfile.txt
drwxr-x--- 9 vikasini vikasini 4096 Sep 2 03:27 ..
-rwxr--r-- 1 vikasini vikasini 0 Sep 2 03:47 permission.txt
drwxr-xr-x 2 vikasini vikasini 4096 Sep 2 03:47 .
vikasini@LAPTOP-DPLUR7N6: ~$ chmod r+w+x permission.txt
chmod: invalid mode: 'r+w+x'
Try 'chmod --help' for more information.
vikasini@LAPTOP-DPLUR7N6: ~$ chmod u+r,g+w,o+x permission.txt
vikasini@LAPTOP-DPLUR7N6: ~$ ls -lart
total 16
-rw-r--r-- 1 vikasini vikasini 0 Sep 1 15:48 copyday1.txt
-rw-r--r-- 1 vikasini vikasini 44 Sep 2 03:21 day1.txt
-rw-r--r-- 1 vikasini vikasini 75 Sep 2 03:27 myfile.txt
drwxr-x--- 9 vikasini vikasini 4096 Sep 2 03:27 ..
-rwxr--r-- 1 vikasini vikasini 0 Sep 2 03:47 permission.txt
drwxr-xr-x 2 vikasini vikasini 4096 Sep 2 03:47 .
vikasini@LAPTOP-DPLUR7N6: ~$
```

```
vikasini@LAPTOP-DPLUR7N6: ~$ chmod 777 permission.txt
vikasini@LAPTOP-DPLUR7N6: ~$ ls -lart
total 16
-rw-r--r-- 1 vikasini vikasini 0 Sep 1 15:48 copyday1.txt
-rw-r--r-- 1 vikasini vikasini 44 Sep 2 03:21 day1.txt
-rw-r--r-- 1 vikasini vikasini 75 Sep 2 03:27 myfile.txt
drwxr-x--- 9 vikasini vikasini 4096 Sep 2 03:27 ..
-rwxrwxrwx 1 vikasini vikasini 0 Sep 2 03:47 permission.txt
drwxr-xr-x 2 vikasini vikasini 4096 Sep 2 03:47 .
vikasini@LAPTOP-DPLUR7N6: ~$ chmod 770 permission.txt
vikasini@LAPTOP-DPLUR7N6: ~$ ls -lart
total 16
-rw-r--r-- 1 vikasini vikasini 0 Sep 1 15:48 copyday1.txt
-rw-r--r-- 1 vikasini vikasini 44 Sep 2 03:21 day1.txt
-rw-r--r-- 1 vikasini vikasini 75 Sep 2 03:27 myfile.txt
drwxr-x--- 9 vikasini vikasini 4096 Sep 2 03:27 ..
-rwxrwx--- 1 vikasini vikasini 0 Sep 2 03:47 permission.txt
drwxr-xr-x 2 vikasini vikasini 4096 Sep 2 03:47 .
vikasini@LAPTOP-DPLUR7N6: ~$ chmod 744 permission.txt
vikasini@LAPTOP-DPLUR7N6: ~$ ls -lart
total 16
-rw-r--r-- 1 vikasini vikasini 0 Sep 1 15:48 copyday1.txt
-rw-r--r-- 1 vikasini vikasini 44 Sep 2 03:21 day1.txt
-rw-r--r-- 1 vikasini vikasini 75 Sep 2 03:27 myfile.txt
drwxr-x--- 9 vikasini vikasini 4096 Sep 2 03:27 ..
-rwxr--r-- 1 vikasini vikasini 0 Sep 2 03:47 permission.txt
drwxr-xr-x 2 vikasini vikasini 4096 Sep 2 03:47 .
vikasini@LAPTOP-DPLUR7N6: ~$ chmod g
chmod: missing operand after 'g'
Try 'chmod --help' for more information.
vikasini@LAPTOP-DPLUR7N6: ~$ chmod g
chmod: missing operand after 'g'
Try 'chmod --help' for more information.
vikasini@LAPTOP-DPLUR7N6: ~$ chmod g+w permission.txt
vikasini@LAPTOP-DPLUR7N6: ~$ ls -lart
total 16
-rw-r--r-- 1 vikasini vikasini 0 Sep 1 15:48 copyday1.txt
```

```
vikasini@LAPTOP-DPLUR7N6: ~$ ls -lart
total 68
-rw-r--r-- 1 vikasini vikasini 3771 Jul 26 07:45 .bashrc
-rw-r--r-- 1 vikasini vikasini 807 Jul 26 07:45 .profile
-rw-r--r-- 1 vikasini vikasini 220 Jul 26 07:45 .bash_logout
drwx----- 2 vikasini vikasini 4096 Jul 26 07:45 .cache
-rw-r--r-- 1 vikasini vikasini 0 Jul 26 08:39 .sudo_as_admin_successful
-rw-r--r-- 1 vikasini vikasini 63 Jul 26 08:46 .gitconfig
drwxr-xr-x 3 vikasini vikasini 4096 Jul 26 11:13 ubuntupractice
drwxr-xr-x 2 vikasini vikasini 4096 Jul 29 16:18 new-folder
drwxr-xr-x 2 vikasini vikasini 4096 Aug 3 16:23 chmod_practice
drwxr-xr-x 5 root root 4096 Aug 29 18:09 ..
-rw----- 1 vikasini vikasini 4558 Sep 1 16:05 .bash_history
-rw-rw-r-- 1 vikasini vikasini 0 Sep 2 03:02 .motd_shown
drwxr-xr-x 2 vikasini vikasini 4096 Sep 2 03:10 practice2
-rw----- 1 vikasini vikasini 20 Sep 2 03:16 .lesshst
drwxr-xr-x 3 vikasini vikasini 4096 Sep 2 03:19 .local
drwxr-x--- 9 vikasini vikasini 4096 Sep 2 03:27 .
drwxr-xr-x 2 vikasini vikasini 4096 Sep 2 03:27 practice
-rw----- 1 vikasini vikasini 882 Sep 2 03:27 .viminfo
vikasini@LAPTOP-DPLUR7N6: ~$ cd ~/practice
vikasini@LAPTOP-DPLUR7N6: ~/practice$ pwd
/home/vikasini/practice
vikasini@LAPTOP-DPLUR7N6: ~/practice$ ls -lart
total 16
-rw-r--r-- 1 vikasini vikasini 0 Sep 1 15:48 copyday1.txt
-rw-r--r-- 1 vikasini vikasini 44 Sep 2 03:21 day1.txt
-rw-r--r-- 1 vikasini vikasini 75 Sep 2 03:27 myfile.txt
drwxr-x--- 9 vikasini vikasini 4096 Sep 2 03:27 ..
drwxr-xr-x 2 vikasini vikasini 4096 Sep 2 03:27 .
vikasini@LAPTOP-DPLUR7N6: ~/practice$ chmod 700
chmod: missing operand after '700'
Try 'chmod --help' for more information.
vikasini@LAPTOP-DPLUR7N6: ~/practice$ touch permission.txt
vikasini@LAPTOP-DPLUR7N6: ~/practice$ ls -l permission.txt
-rw-r--r-- 1 vikasini vikasini 0 Sep 2 03:47 permission.txt
vikasini@LAPTOP-DPLUR7N6: ~/practice$ chmod 777 permission.txt
```

```
vikasini@LAPTOP-DPLUR7N6: X + v
vikasini@LAPTOP-DPLUR7N6:~/practice$ chmod 777 permission.txt
vikasini@LAPTOP-DPLUR7N6:~/practice$ ls -lart
total 16
-rw-r--r-- 1 vikasini vikasini 0 Sep 1 15:48 copyday1.txt
-rw-r--r-- 1 vikasini vikasini 44 Sep 2 03:21 day1.txt
-rw-r--r-- 1 vikasini vikasini 75 Sep 2 03:27 myfile.txt
drwxr-x--- 9 vikasini vikasini 4096 Sep 2 03:27 ..
-rwxrwxrwx 1 vikasini vikasini 0 Sep 2 03:47 permission.txt
drwxr-xr-x 2 vikasini vikasini 4096 Sep 2 03:47 .
vikasini@LAPTOP-DPLUR7N6:~/practice$ chmod 770 permission.txt
vikasini@LAPTOP-DPLUR7N6:~/practice$ ls -lart
total 16
-rw-r--r-- 1 vikasini vikasini 0 Sep 1 15:48 copyday1.txt
-rw-r--r-- 1 vikasini vikasini 44 Sep 2 03:21 day1.txt
-rw-r--r-- 1 vikasini vikasini 75 Sep 2 03:27 myfile.txt
drwxr-x--- 9 vikasini vikasini 4096 Sep 2 03:27 ..
-rwxrwx--- 1 vikasini vikasini 0 Sep 2 03:47 permission.txt
drwxr-xr-x 2 vikasini vikasini 4096 Sep 2 03:47 .
vikasini@LAPTOP-DPLUR7N6:~/practice$ chmod 744 permission.txt
vikasini@LAPTOP-DPLUR7N6:~/practice$ ls -lart
total 16
-rw-r--r-- 1 vikasini vikasini 0 Sep 1 15:48 copyday1.txt
-rw-r--r-- 1 vikasini vikasini 44 Sep 2 03:21 day1.txt
-rw-r--r-- 1 vikasini vikasini 75 Sep 2 03:27 myfile.txt
drwxr-x--- 9 vikasini vikasini 4096 Sep 2 03:27 ..
-rwxr--r-- 1 vikasini vikasini 0 Sep 2 03:47 permission.txt
drwxr-xr-x 2 vikasini vikasini 4096 Sep 2 03:47 .
vikasini@LAPTOP-DPLUR7N6:~/practice$ chmod g
chmod: missing operand after 'g'
Try 'chmod --help' for more information.
vikasini@LAPTOP-DPLUR7N6:~/practice$ chmod g
chmod: missing operand after 'g'
Try 'chmod --help' for more information.
vikasini@LAPTOP-DPLUR7N6:~/practice$ chmod g+w permission.txt
vikasini@LAPTOP-DPLUR7N6:~/practice$ ls -lart
total 16
-rw-r--r-- 1 vikasini vikasini 0 Sep 1 15:48 copyday1.txt

vikasini@LAPTOP-DPLUR7N6: X + v
vikasini@LAPTOP-DPLUR7N6:~$ sudo useradd -m vikash
[sudo] password for vikasini:
vikasini@LAPTOP-DPLUR7N6:~$ id vikash
uid=1003(vikash) gid=1003(vikash) groups=1003(vikash)
vikasini@LAPTOP-DPLUR7N6:~$ sudo passwd vikash
New password:
Retype new password:
passwd: password updated successfully
vikasini@LAPTOP-DPLUR7N6:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:./nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:./usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:./usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101:./nonexistent:/usr/sbin/nologin
syslog:x:102:102:./nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:./usr/sbin/nologin
uidd:x:103:103:./run/uidd:/usr/sbin/nologin
landscape:x:104:105:./var/lib/landscape:/usr/sbin/nologin
polkitd:x:990:990:User for polkitd:./usr/sbin/nologin
vikasini:x:1000:1000:./home/vikasini:/bin/bash
```



```
vikasini@LAPTOP-DPLUR7N6:~$ grep vikash /etc/passwd
vikash:x:1003:1003::/home/vikash:/bin/sh
vikasini@LAPTOP-DPLUR7N6:~$ whoami
vikasini
vikasini@LAPTOP-DPLUR7N6:~$ sudo -i
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.18.33.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep  2 16:25:26 UTC 2026

System load:  0.17               Processes:    38
Usage of /:   0.2% of 1006.85GB  Users logged in: 1
Memory usage: 4%                IPv4 address for eth0: 172.17.239.204
Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

https://ubuntu.com/workshop

This message is shown once a day. To disable it please create the
/root/.hushlogin file.
root@LAPTOP-DPLUR7N6:~# whoami
root
root@LAPTOP-DPLUR7N6:~# cd /
root@LAPTOP-DPLUR7N6:/# ls
bin      etc      lib.usr-is-merged  mnt      run      srv      var      wslOMPnFn  wslhOPmEn
bin.usr-is-merged  home     lib64              opt      sbin     sys      wslBlcbgj  wslbopKE  wsljFNeDF
boot      init     lost+found         proc     sbin.usr-is-merged  tmp      wslICikEF  wslmGLBn  wslkDGaFF
dev      lib      media              root     snap     usr      wslNfioFC  wslkOLBF

root@LAPTOP-DPLUR7N6:/# cd /root
root@LAPTOP-DPLUR7N6:~# ls
root@LAPTOP-DPLUR7N6:~# exit
logout

vikasini@LAPTOP-DPLUR7N6:~$ sudo userdel -r vikash
userdel: vikash mail spool (/var/mail/vikash) not found
vikasini@LAPTOP-DPLUR7N6:~$ id vikash
id: 'vikash': no such user
vikasini@LAPTOP-DPLUR7N6:~$ sudo useradd -m vikash
vikasini@LAPTOP-DPLUR7N6:~$ sudo passwd vikash
New password:
Retype new password:
passwd: password updated successfully
vikasini@LAPTOP-DPLUR7N6:~$ sudo groupadd vishu
vikasini@LAPTOP-DPLUR7N6:~$ cat /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,vikasini
tty:x:5:
disk:x:6:
lp:x:7:
mail:x:8:
news:x:9:
uucp:x:10:
man:x:12:
proxy:x:13:
kmem:x:15:
dialout:x:20:vikasini
fax:x:21:
voice:x:22:
cdrom:x:24:vikasini
floppy:x:25:vikasini
tape:x:26:
sudo:x:27:vikasini
audio:x:29:vikasini
dip:x:30:vikasini
www-data:x:33:
backup:x:34:
operator:x:37:
```

```
vikasini@LAPTOP-DPLUR7N6:~$ sudo useradd -m vikasini2
useradd: user 'vikasini2' already exists
vikasini@LAPTOP-DPLUR7N6:~$ id
uid=1000(vikasini) gid=1000(vikasini) groups=1000(vikasini),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev),100(users),107(netdev)
vikasini@LAPTOP-DPLUR7N6:~$ id vikasini2
uid=1004(vikasini2) gid=1005(vikasini2) groups=1005(vikasini2)
vikasini@LAPTOP-DPLUR7N6:~$ sudo usermod -d /home/..... -m vikasini2
usermod: The previous home directory (/home/vikasini2) does not exist or is inaccessible. Move cannot be completed.
vikasini@LAPTOP-DPLUR7N6:~$ getent passwd vikasini2
vikasini2:x:1004:1005::/home/.....:/bin/sh
vikasini@LAPTOP-DPLUR7N6:~$ sudo mkdir /home/newhome
vikasini@LAPTOP-DPLUR7N6:~$ sudo usermod -d /home/newhome -m vikasini2
usermod: directory /home/newhome exists
vikasini@LAPTOP-DPLUR7N6:~$ getent passwd vikasini2
vikasini2:x:1004:1005::/home/newhome:/bin/sh
vikasini@LAPTOP-DPLUR7N6:~$ cd /home/vikasini
vikasini@LAPTOP-DPLUR7N6:~$ pwd
/home/vikasini
vikasini@LAPTOP-DPLUR7N6:~$ vim vikash-info.txt
vikasini@LAPTOP-DPLUR7N6:~$ cat vikash-info.txt
Hello vikash
How are you
vikasini@LAPTOP-DPLUR7N6:~$ echo "This is my linux practice" >> vikash-info.txt
vikasini@LAPTOP-DPLUR7N6:~$ cat vikash-info.txt
Hello vikash
How are you
This is my linux practice
vikasini@LAPTOP-DPLUR7N6:~$ mv test.txt vikash.txt
mv: cannot stat 'test.txt': No such file or directory
vikasini@LAPTOP-DPLUR7N6:~$ touch text.txt
vikasini@LAPTOP-DPLUR7N6:~$ ls
chmod_practice new-folder practice practice2 text.txt ubuntupractice vikash-info.txt
vikasini@LAPTOP-DPLUR7N6:~$ mv text.txt vikash.txt
vikasini@LAPTOP-DPLUR7N6:~$ ls
chmod_practice new-folder practice practice2 ubuntupractice vikash-info.txt vikash.txt
vikasini@LAPTOP-DPLUR7N6:~$
```

CONCLUSION:

Overall, this practical helped me gain basic knowledge and confidence in using Linux commands and understand how Linux manages files, users, permissions, groups, and processes.